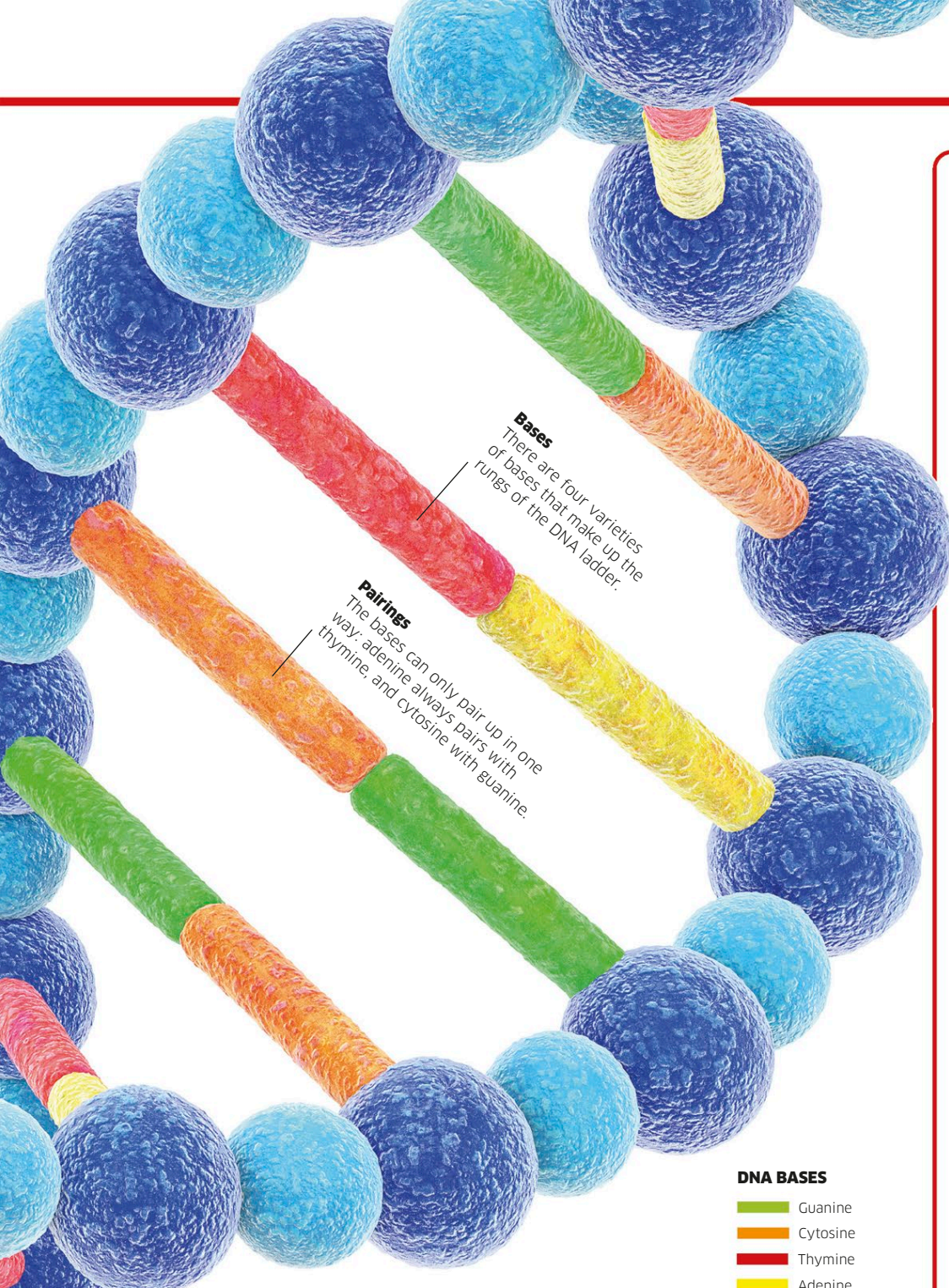
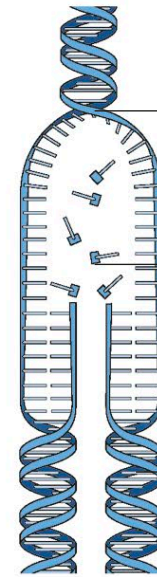




## DNA replication

Cells replicate themselves by splitting in two. Therefore, all the instructions held in DNA must be copied before a cell divides, so each new cell will have a full set. The DNA does this by splitting into two strands. Each of these then provides a template for building a new double helix.



**1** Each molecule of DNA is made of two complementary strands. When it is ready to replicate, the double helix unzips into two separate strands.

**2** New DNA building blocks come together to make the other sides of each double helix. As each base can only pair with one other, it is clear which blocks are needed to complete the "ladder."

**3** Two new double helices are formed. Each of these is then ready to go into two new cells when the original cell divides.

## What gets inherited?

Many human features, such as eye color, hair color, and blood type, are due to particular genes. Different varieties of genes, called alleles, determine variation in these characteristics. Other characteristics, such as height, are affected by many genes working together, but also by other factors, such as diet.

### Genes

Some characteristics are only inherited from parents.



### Genes and environment

Other characteristics are influenced by genes and the environment.



## How inheritance works

Most organisms have two of each kind of gene—one from each parent. Many genes have two or more variations, called alleles, so the genes an animal inherits from its mother and father may be identical or different. When two animals, such as rabbits, reproduce, there are many different combinations of alleles their offspring can receive. Some alleles are dominant (like those for brown fur), and when a baby rabbit has two different alleles it will have the characteristic of the dominant allele. Other alleles are recessive, and babies will only have the characteristic they determine if they have two of them. This explains why some children inherit physical characteristics not seen in their parents.

